

COMPUTER SCIENCE TRIPOS Part IB – 2023 – Paper 7

3 Economics, Law and Ethics (ah793)

philosophies of
ethics

- (a) Describe the six branches of philosophy, explaining what insight they might give into technology ethics. [12 marks]

Answer: *Metaphysics* is concerned with the study of reality. Relevant concepts include misinformation (or false information) and disinformation (the deliberate spread of misinformation). Practically, we apply metaphysics when we ask what is reality when it is virtual or augmented?

Epistemology explores knowledge, the methods of acquiring it, and its limitations. To answer the metaphysical question ‘Can machines think?’, we need to ask epistemological questions about how we can know if machines think, and who or what our sources of knowledge are. When it comes to AI, can results be reproduced? How do we know how AI, especially ‘deep learning’ reached a conclusion? Should information be free? Or censored? What about mis and disinformation? Or hate speech? Who creates knowledge, and how is it established, tested and verified?

Political philosophy explores the relationship between individuals and society. It asks what sort of society do we want to live in? How should it be organised? Who should rule? What are people’s rights and responsibilities? In relation to technology, large technology companies are effectively political actors at the national and international level due to their power, wealth, number of customers, and market dominance. In some cases, their platforms have the ability to directly shape political discourse.

Logic is used to determine if an argument is sound or if a hypothesis is false. The Turing test is designed to answer the question ‘can machines think?’ CAPTCHAs are a type of Turing test, aimed at differentiating between humans and bots.

Aesthetics relates to our senses and perception. It is a concern when it comes to the creation and use of technology and tools. The user interface and user experience can hook users into online platforms, to the extreme this can lead to addiction. Aesthetics is also important when it comes to considering accessibility and inclusivity, for example, meeting the design needs of users with low vision, users with screen readers, users who are deaf or hard of hearing, with physical or motor disabilities, users with dyslexia, or autistic people.

Ethics is concerned with right and wrong, how we should live and what constitutes a good life. Technology ethics is an area of applied ethics, or ethics put into practice. Ethics is not just a checklist, that can be done once and forgotten about. It should be a continued conversation, as dynamic as the technology it is applied to. Number of different philosophies of ethics, or frameworks to view ethical issues.

(For each branch of philosophy, one mark for a good description, and a further mark for a relevant and well-explained application to questions of technology ethics).

philosophies of
ethics

- (b) Using a consequentialist approach to the philosophy of ethics, outline the ethical implications of facial recognition technology. Your answer should identify the potential benefits and harms of using the technology. [8 marks]

Answer: Consequentialism is based on two principles: 1. Whether an act is right or wrong depends only on the results of that act – i.e. the end justifies the means; and 2. The more good consequences an act produces, the better or more right that act (2 marks). In the context of facial recognition technology, a consequentialist approach would focus on the potential harms and benefits that the technology may cause.

Potential benefits of facial recognition technology potentially include increased security (e.g. identify people who should not be in secured areas) and convenience (granting access to people who do have authorisation to access secured areas) (2 marks).

— *Solution notes* —

However, facial recognition technology also raises potential harms. One concern is privacy, as the technology relies on the collection and storage of large amounts of personal data, which can be used for surveillance, tracking, and profiling individuals without their knowledge or consent (*2 marks*). Another concern is accuracy, as facial recognition technology can produce false positives and false negatives, which can lead to wrongful identification, and can be used to justify discrimination or repression (*2 marks*).
